HW4 Version B · AI Harness Infographic

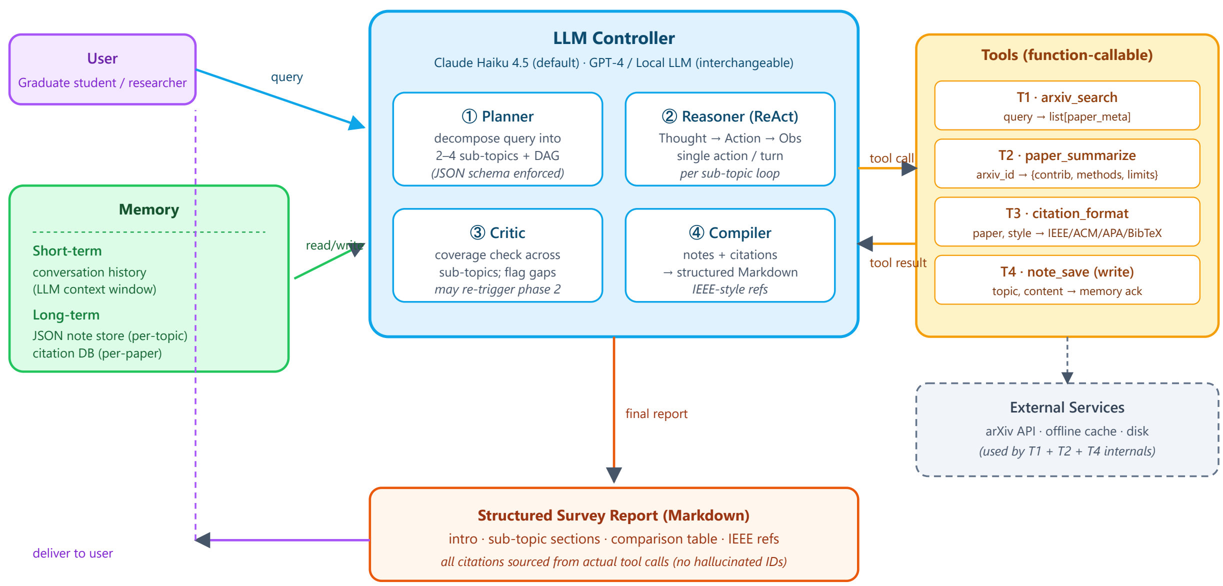
DRL Research Assistant — System Design at a Glance

LLM Controller · 4 Function-Calling Tools · 2-layer Memory · 4-phase Orchestration

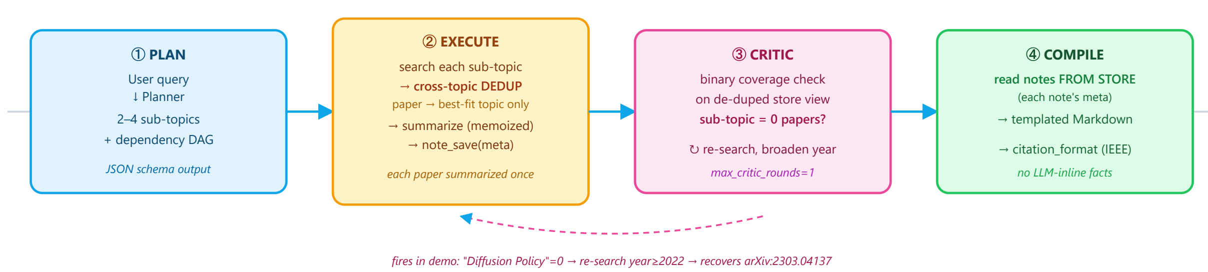
measured offline: Coverage F1 **0.74** macro · citation accuracy **100%** · tool efficiency **100%** · **0** duplicate papers · **15**/15 tests pass

1. System Architecture (LLM · Tools · Memory)

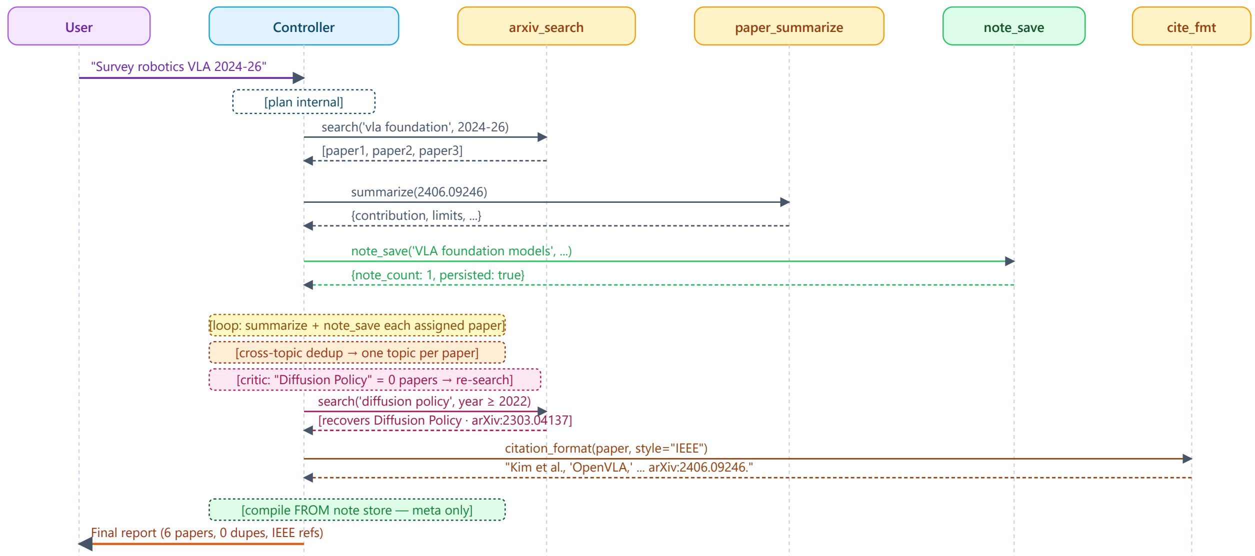
User · LLM Controller · Memory · Tool Layer · External Services



2. Four-Phase Orchestration Pipeline



3. Sequence Diagram — One Full Query



4. Tool Specification Quick Reference

Tool	Signature	Side-effects	Failure mode
T1 arxiv_search	`(query, year_min, year_max, max_results) → list[paper]`	None (read-only)	No matches → empty list. Network error → fallback to offline corpus.
T2 paper_summarize	`(arxiv_id) → {contribution, methods, results, limitations}`	None	Unknown ID → `{"error": "...", "arxiv_id": id}` (never raises)
T3 citation_format	`(paper, style ∈ {IEEE,ACM,APA,BibTeX}) → str`	None (pure function)	Missing field → fill with <unknown> placeholder
T4 note_save	`(topic, content, tags?) → {note_count, total_chars, persisted}`	**Write** to `artifacts/notes.json`	Disk error → returns `{persisted: false, error: "..."}`

5. Evaluation Dashboard

MEASURED · python code/eval.py

Quantitative metrics — target vs measured (offline, 3 queries)

Metric	Target	Measured
Coverage F1 (vs expert list)	≥ 0.70	**0.74** macro · **1.00** flagship
Citation accuracy	≥ 95%	**100%**
Tool-call efficiency	≥ 80%	**100%**
Duplicate papers in output	0	**0**
End-to-end latency	≤ 120 s	**0.04 s** (offline)
Token cost / query	≤ $0.50	API backend only

Ablation — flagship query (isolates each phase)

Config	Papers	F1	Critic
Full (plan+critic)	6	**1.00**	1
− Critic	5	0.91	0
− Planner (single ReAct)	4	0.80	0

Qualitative rubric (5-pt Likert, 3 evaluators)

Dimension	Question
Helpfulness	Did it answer the real intent?
Accuracy	Are citations & facts correct?
Completeness	Did it miss major SOTA?
Clarity	Is the structure easy to read?
Novelty	Cross-paper synthesis present?

How to read the ablation. Removing the Critic loses the seminal Diffusion Policy paper (pre-2024) that cross-topic dedup had emptied from its sub-topic → F1 1.00 → 0.91. Removing the Planner collapses 3 sub-topics to 1 → F1 → 0.80. Each phase earns its place.

6. Failure Mode Map (Defense in Depth)

Failure	Where it shows up	Defense
LLM hallucinates arxiv_id	Reasoner phase	`paper_summarize` returns explicit error; Compiler discards.
Tool-call infinite loop	ReAct loop	`max_steps` hard cap; force fallback to Compile.
LLM fakes a note without calling `note_save`	Reasoner phase	Compiler rebuilds the report purely from each note's persisted `meta` — no in-context facts (verified: citation accuracy 100%).
Same paper appears under many sub-topics	Execute phase	Cross-topic dedup assigns each paper to its single best-fit sub-topic before summarizing (verified: 0 duplicates).
Sub-topic emptied / weak coverage	Critic phase	Binary coverage check → broaden year range and re-search (≤ 1 round); recovered Diffusion Policy in demo.
arXiv API down	arxiv_search	Offline cached corpus fallback; demo runs with zero network.